LAB PGM – 5 AES Algorithm

```
import java.util.*;
import javax.crypto.*;
import javax.xml.bind.DatatypeConverter;

public class lab5AES {
    public static void main(String[] args) throws Exception {

        Scanner sc = new Scanner(System.in);

        KeyGenerator kg = KeyGenerator.getInstance("AES");
        kg.init(128);
        SecretKey key = kg.generateKey();

        Cipher c = Cipher.getInstance("AES");

        System.out.print("Enter message: ");
        String msg = sc.nextLine();

        c.init(Cipher.ENCRYPT_MODE, key);
        byte[] encrypted = c.doFinal(msg.getBytes());
```

```
        String code = DatatypeConverter.printBase64Binary(encrypted);


        c.init(Cipher.DECRYPT_MODE, key);
        byte[] decrypted =
c.doFinal(DatatypeConverter.parseBase64Binary(code));


        System.out.println("Encrypted message: " + code);
        System.out.println("Decrypted message: " + new String(decrypted, "UTF-
8"));


        sc.close();
    }
}
```